

SHU-HAO WU

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SUMMARY

Experienced in both infrastructure and backend. Skilled in backend API development with Node.js, Java, Python, Ruby, Redis and MySQL. Also, design and operate AWS, GCP, and Kubernetes environments with through IaC.

- **RUGI**
 - Created an unofficial multilingual mobile/PWA guide for RubyKaigi 2026's ~2,000 attendees, using **Next.js** and **Capacitor** to centralize sessions, 42 community events, local recommendations, maps, and offline access.

TECHNICAL SKILLS

Programming Languages & Databases: Python, Java, JavaScript, Ruby, MongoDB, MySQL, PostgreSQL, Redis, SQLite, SQL Server

Frameworks: Django, Express.js, FastAPI, Flask, Node.js, Spring Boot

Technologies: AWS, Azure, CSS, CI/CD, Docker, GCP, Git, HTML, Linux, Power BI, Tableau, Terraform

EDUCATION

University of Illinois at Urbana-Champaign, Champaign, IL, USA

August 2021 – May 2023

- Master of Science in Information Science

Japanese-Language Proficiency Test (JLPT) N2

PROFESSIONAL EXPERIENCE

Treasure Data / Treasure AI

January 2026 – Present

Software Engineer

Tokyo, Japan

- Session Idle Timeout Implementation
 - Built auto session timeout for two **React** apps used by 400+ corporate clients, replacing a third-party library with a custom solution using React hooks and **Redux**
 - Built smart polling system with **SWR** that checks session status more frequently as timeout approaches, reducing unnecessary server calls
 - Synced session state across browser tabs using BroadcastChannel API and **TypeScript**, fixing unexpected logout issues
 - Published a shared npm package to unify API calls across multiple codebases, unblocking the feature release
- API Usage Development
 - Built automated email reminder workflows for expiring API keys, helping customers avoid unexpected API disruptions
 - Designed and implemented **MySQL** database schema changes and backend logic for configurable API key expiration and reminder policies
 - Developed an API key management interface in **React** with configurable expiration settings and notification banners, improving visibility of upcoming API key expirations
 - Led the implementation of API key expiration and reminder mechanisms by updating authentication flows and **Ruby on Rails** backend endpoints, preventing indefinite API key usage

THECOO Inc.

November 2024 – December 2025

Software Engineer

Tokyo, Japan

- AWS Migration Proof of Concept

- Audited existing **GCP** infrastructure and designed a modular **AWS** architecture using **Terraform** with reusable layers of core infrastructure
- Documented and shared migration strategy across teams to support future environment expansion and cross-cloud architecture planning
- Evaluated ArgoCD vs Terraform for managing **Kubernetes** state, focusing on GitOps principles and environment consistency
- Initiated and led PoC project to evaluate migrating services from GCP to AWS to optimize long-term infrastructure costs
- Force Logout Functionality
 - Designed custom **Redis** session store by using **Node.js** to support user ID-based session deletion, enabling force logout feature to remove all user sessions across devices when triggered by account deletion or password reset
 - Integrated force logout mechanism into key API routes (e.g., password reset, account deletion), ensuring session invalidation is correctly triggered
 - Wrote **Jest** integration tests to ensure reliability of the custom **Redis** session store, covering session creation, retrieval, deletion, and user-based force logout
- GCP Environment Separation
 - Implemented Private Service Connect (PSC) to securely connect environments without exposing internal IPs
 - Isolated production and non-production environments in our GCP infrastructure by refactoring backend **Node.js** configuration and **Terraform** infrastructure code; migrated key resources including **Redis**, **BigQuery**, etc, improving data access control, deployment safety, and environment-specific debugging efficiency
 - Refactored backend Node.js configurations and Terraform modules for multi-environment compatibility
- Operation Fee Implementation
 - Designed and implemented a new "operation fee" system for scratch-based games using points, generating ~¥170 million in net additional revenue annually
 - Migrated **MySQL** schema by creating a new operation fee table and adding foreign key to existing tables such as lottery, ensuring structured fee management
 - Ensured accounting accuracy by updating admin-side sales reports and excluding operation fees from royalties and payouts
 - Wrote unit and integration tests using **Jest** to validate fee calculation logic and ensure correctness across backend and reporting systems
- Slack Bot for Document Retrieval
 - Developed **Slack bot** by using **Node.js** to assist employees and HR in retrieving relevant documents, reduce search time from 30 minutes to 5 seconds
 - Implemented **Slack API** calls to fetch messages from channels based on user queries
 - Used **Google Drive API** to list and filter documents, extracting filenames and links to display in Slack
- Terraform Infrastructure Standardization (AWS)
 - Fixed resource inconsistencies in **AWS Lambda** by verifying dependencies with services like **EventBridge**, ensuring all related resources were properly imported into Terraform
 - Imported manually created resources into Terraform and upgraded provider versions to eliminate deprecated syntax
 - Refactored **Kubernetes** environment from MVP by restructuring configurations and optimizing resource allocation improving deployment stability, reducing setup time

Allstate Insurance Company

Data Engineer

July 2023 – March 2024

Remote, USA

- Architected comprehensive data pipeline using **Azure Data Factory**, reducing data processing time